

SARANSH SURANA

+1 631-816-8344 | ssurana818@gmail.com | linkedin.com/in/saransh-surana | saransh-surana | Github

EDUCATION

Stony Brook University | M.S. Data Science | Stony Brook, NY

Aug 2023 – May 2025

- **Research Assistant (Jan 2025 – May 2025):** Built **Spark** + Python pipelines for large unstructured PDF corpora, performing OCR, text cleaning, and **distributed** preprocessing for downstream extraction tasks.

WORK EXPERIENCE

Schizophrenia & Psychosis Action Alliance

Jul 2025 – Present

AI Software Engineer

Remote, USA

- Benchmarked **GPT-4**, **Claude 3.5**, **Gemini Flash**, and **DeepSeek V2** on a 250-sample extraction suite for behavioral health data, evaluating JSON validity and unsupported generation rates to select **GPT-4** for production.
- Built an end-to-end **LLM** service extraction pipeline using **prompt engineering**, **few-shot** examples, citation grounding, and **JSON Schema** validation, improving parse success from 78% to 96% and reducing unsupported generations by 25%.
- Ran 50+ versioned experiments in **MLflow** across model choice, prompt variants, chunking, and retrieval settings for regression tracking.
- Implemented cloud-native microservices using **FastAPI** with async I/O, **Celery** workers, and **Redis** queues, to run a production distributed LLM extraction pipeline, processing 1,200–2,000 records per hour with under 2% timeouts.
- Shipped a **React.js** + **TypeScript** directory UI with search, facet filters (county, service tags), and paginated tables.
- Set up **Docker** builds and **GitHub Actions** CI with ruff and pytest gates, using **GitHub Copilot** for test scaffolding and deploying on **Azure App Service** with environment config and secrets.

Ford Motor Company

May 2024 – Aug 2024

Data Science Intern

Dearborn, MI, USA

- Queried and manipulated large datasets using Python and SQL to support ML pipelines and analysis on GCP.
- Performed **EDA** on waveform sensor data, then built and evaluated anomaly detection models using **Isolation Forest** and **One-Class SVM** with **PCA** and **Bayesian hyperparameter** tuning, achieving 78% recall and 73% precision.
- Built a **Vertex AI RAG system** over 100,000+ warranty conversations, improving retrieval of failure trends and root-cause patterns for engineer investigation.

Napuor

Aug 2022 – Jan 2023

Data Science Intern

Bengaluru, KA, India

- Trained, productionized, and deployed **time-series demand forecasting** and inventory optimization models with **XGBoost** on **Azure**, reducing forecast error by 18%.
- Tracked experiments in **MLflow** and monitored runs with **Prometheus** and **Grafana** dashboards.

OPEN SOURCE CONTRIBUTIONS

- **Haystack Core Integrations** (deepset-ai): 5 merged PRs, added **async run** for **LlamaCppChatGenerator** with async tests, added **Jina** integration tests, added **SUPPORTED MODELS** to **AnthropicVertexChatGenerator**, fixed **llama-stack** defaults, and simplified CI to reduce flakes.
- **Outlines** (dottxt-ai), **Skrub**, **Statsmodels**: 3 merged contributions, added parametrized tokenizer smoke tests across 10+ Transformers tokenizers plus API and doc fixes, improving CI coverage and reducing user-facing ambiguity.
- **Outlines** (dottxt-ai): Reported **LlamaCppTokenizer** bugs affecting **EOS attention masking** and fallback **vocabulary extraction**, with a minimal reproducible script for maintainers.

PROJECTS

Reinforcement Learning Book Recommender | *Python, PyTorch, Stable-Baselines3, Gymnasium, Pandas, Tensorflow*

- Trained a PPO agent on 10K+ interactions, achieving a 19% higher cumulative reward than BiLSTM and logistic baselines with detailed error analysis.
- Formulated the RL problem and tuned policy-gradient hyperparameters for stable, higher-reward policies.

Agentic AI Workflow for Member Support and Risk Ops | *Python, LangGraph, FastAPI, OpenAI API, PostgreSQL*

- Built an agent-based workflow for 3,500+ member support cases across triage, retrieval, and response drafting.
- Benchmarked routing and validation agents, improving response accuracy by 14% and helping flag higher-risk cases faster.

SKILLS

- **Languages:** Python, SQL, C/C++, TypeScript, Java, C#, Go
- **AI and LLMs:** PyTorch, scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph, GPT-4, Claude, Gemini
- **Retrieval:** Sentence-Transformers, FAISS, RAG, chunking, hybrid search, reranking
- **Backend and Infrastructure:** FastAPI, REST APIs, Celery, Redis, Docker, Azure, Kubernetes
- **Data and MLOps:** Spark, BigQuery, MLflow, Prometheus, Grafana, GitHub Actions